

3.1 Data Structures in Python

ڈیٹا سٹرکچر (Data Structure) پروگرام میں ڈیٹا کو ترتیب دینے کا طریقہ ہے۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 01 of 17

3.1 Data Structures in Python

LEARN IT

A **data structure** is a way of organising values in a program so you can store them, find them and change them. Every useful program is built on data structures. Python gives you two layers:

- **Primitive data structures** — the basic types that hold **one** value. Examples: integers (`int`), floating-point numbers (`float`), and booleans (`bool`).
- **Abstract data structures** — higher-level containers built from primitive types so you can hold **many** values and run richer operations. Examples: **lists, tuples, sets, dictionaries**.

Python has these four abstract structures built in. You do not install anything extra. That is why collecting marks, names or settings is simpler in Python than in many other languages.

Category	Description	Examples
Primitive	Basic types that represent a single value.	<code>int</code> , <code>float</code> , <code>bool</code>
Abstract	Higher-level structures for collections and complex operations.	Lists, tuples, sets, dictionaries

WORKED EXAMPLE

One value versus a collection. A student's age is one integer. That student's three test marks need a list.

```
age = 17
height = 5.8
passed = True
marks = [80, 75, 90]
print(type(age))
print(type(height))
print(type(passed))
print(type(marks))
```

OUTPUT

```
<class 'int'>
<class 'float'>
<class 'bool'>
<class 'list'>
```

`type()` tells you which structure a value has. `age` is a primitive integer. `marks` is an abstract list that holds three integers together.

REMEMBER IT

Primitive = one piece. Abstract = a box of pieces. Think of a single rupee coin (primitive) versus a wallet that can hold many coins (list, tuple, set or dictionary).

COMMON MISTAKE

Do not call a list a “variable type” and stop there. A list is a **structure**: it has an order, methods, and rules. The four abstract structures look similar at first (they all hold several values) but they follow different rules. Learn those rules one by one in 3.1.1–3.1.4.

اردو میں سمجھیں

ڈیٹا سٹرکچر (Data Structure) پروگرام میں ڈیٹا کو ترتیب دینے کا طریقہ ہے۔ پرائمیٹو اقسام جیسے عدد (int) صرف ایک قیمت رکھتی ہیں، جبکہ فہرست، ٹیپل، سیٹ اور ڈکشنری کئی قیمتیں ایک ساتھ سنبھالتی ہیں۔ پائتھن میں یہ چاروں سٹرکچرز پہلے سے موجود ہیں، اس لیے ڈیٹا جمع کرنا دوسری زبانوں کے مقابلے میں آسان ہے۔

CHECK YOURSELF

1. Name two primitive types and two abstract data structures in Python.
2. Why is a list called an abstract data structure rather than a primitive type?

Answers: 1. Primitive: any two of int, float, bool. Abstract: any two of list, tuple, set, dictionary.
2. Because it is a higher-level container built from primitive values, holding many items and supporting operations such as append and slice.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Python built-in structures in this chapter include:

- a. list, tuple, set, dict
- b. only arrays of C
- c. only trees
- d. only graphs

MCQ 2. You choose among them by:

- a. Order? Duplicates? Mutability? Key lookup?
- b. The logo
- c. The font
- d. The sprint

MCQ 3. Mutable means:

- a. You can change it in place
- b. Frozen forever
- c. Always sorted
- d. Always unique

MCQ 4. A dict maps:

- a. Keys to values
- b. Only numbers to NAND
- c. Only OSI layers
- d. Only files

MCQ 5. A set is best for:

- a. Unique items / membership tests
- b. Ordered ranking with duplicates
- c. LIFO
- d. FIFO

MCQ 6. This lecture is the map; later lectures:

- a. Zoom into each type
- b. Delete Python
- c. Draw only ER
- d. Draw only Figma

Extra short questions

1. One-line each: list, tuple, set, dict.
2. Which for student names that may repeat seats?
3. Which for a record of roll→marks?
4. Which as a function return of (x,y) that must not change?

Extra long question

1. Four college needs → choose the Python DS and justify with one operation each.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. List: ordered mutable sequence. Tuple: ordered immutable. Set: unique unordered. Dict: key→value.
2. List (order). If unique roll numbers: set or dict.
3. Dict.
4. Tuple.

Long

1. Attendance ordered: list. RGB colour constant: tuple. Club unique IDs: set. Phonebook: dict. Show one line of code each.

Next lecture: 3.1.1 Lists (Ordered, Mutable, Methods)